	Faculty of Engineering **Stellenbosch University** ***Fakulteit Ingenieurswese*** ***Universiteit Stellenbosch***	

Data Analytics 344 ***Data Analitika 344***			**Assessment 1** September 2019 ***Assessering 1*** *September 2019*
Duration ***Tydsduur***	**2 h**	**Full marks** ***Volpunte***	**97**

	Names ***Name***	Signatures to confirm: Question paper and marking scheme are correct and aligned with module outcomes *Handtekeninge ter bevestiging: Vraestel en merkskema is korrek en belyn met module uitkomste*
Examiner(s) ***Eksaminator(e)***	Prof A.P. Engelbrecht	
Moderator(s) ***Moderator(e)***	Prof J.H. Van Vuuren	

Instructions	**Instruksies**
The Engineering Faculty's "General Stipulations for Under- and Postgraduate Modules" apply to this assessment.	Fakulteit Ingenieurswese se "Algemene Bepalings vir Voor- en Nagraadse Modules" is op hierdie assessering van toepassing.
Answer all the questions.	Beantwoord al die vrae
The marks for each question are indicated to the right of the question in square brackets.	Die punte vir elke vraag word regs van die vraag in vierkantige hakies aangedui.
This is a **closed book assessment**. No information sources may be consulted.	Hierdie is 'n **toeboek assessering**. Geen inligtingbronne mag geraadpleeg word nie.
Pocket calculators prescribed for the first two years of the BEng may be used.	**Sakrekenaars** voorgeskryf vir die eerste twee jaar van die BIng mag gebruik word.

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Section A: /*Afdeling A* [30]

This section covers the introduction to data analytics topic of the study material./*Hierdie afdeling dek die inleiding tot die data analitika onderwerp van die studie materiaal.*

Question 1: /*Vraag 1* [2]

(a) You have two data files to analyze. The one data file is in total 2.3EB, and the other file is 1.1ZB. Which one of these data files is the largest?

(b) Give an example of unstructured data.

(a) Jy het twee data leêrs om te ontleed. Die een leêr is in totaal 2.3EB, terwyl die ander leêr 1.1ZB is. Watter een van hierdie data leêrs is die grootste? [1]

(b) Gee 'n voorbeeld van ongestruktureerde data. [1]

Question 2: /*Vraag 2* [12]

The concept of *big data* is best described by four primary characteristics. Name each one of these characteristics, give a concise definition of each, and then discuss in short the implications that each characteristic has for data analytics.

Die *groot data* konsep word die beste beskryf deur vier primêre eienskappe. Noem elkeen van hierdie eienskappe, gee 'n bondige definisie van elk, en bespreek dan in kort die implikasies wat elke eienskap vir data analitika het.

Question 3: /*Vraag 3* [2]

Data science, specifically data analytics, adds another characteristic to the *big data* concept. Name this characteristic and give a concise definition thereof.

Datawetenskap, spesifiek data analitika, voeg nog 'n eienskap by tot die *groot data* konsep. Noem hierdie eienskap en gee 'n bondige definisie daarvan.

Question 4: /*Vraag 4* [3]

Data can be either stationary or non-stationary.

(a) What is the main difference between these types of data?

(b) What are the main implications of non-stationary data on data analytics?

Data kan of stasionêr of nie-stasionêr wees.

(a) Wat is die hoof verskil tussen hierdie twee data tipes? [1]

(b) Wat is die hoof implikasies van nie-stasionêre data vir data analitika? [2]

Question 5: /*Vraag 5* [3]

Enterprise data is usually stored in a data warehouse, a number of data marts, and/or in a data lake. Describe in short the main differences between these three data store types.

Ondernemingsdata word gewoonlik in 'n data pakhuis, 'n aantal data markte, en/of in 'n data meer gestoor. Beskryf kortliks die hoof verskille tussen herdie drie tipes data store.

Question 6: /*Vraag 6* **[5]**

(a) Give a concise definition of data analytics.

(b) What is the main difference between visual analytics and data analytics?

(c) Name one aspect that makes image analytics difficult.

(a) Gee 'n bondige definisie van data analitika. [2]

(b) Wat is die hoof verskil tussen visuele analitika en data analitika? [2]

(c) Noem een aspek wat beeld analitika moeilik maak. [1]

Question 7: /*Vraag 7* **[3]**

For each of the following analytics use cases, identify if it is an example of descriptive analytics, diagnostic analytics, predictive analytics, or prescriptive analytics.

(a) From a data set obtained from a health care provider, identify patients who are clinically obese.

(b) For patients identified as clinically obese, propose treatment with reference to diabetes and LDL cholesterol.

(c) A cellular communications provider has experienced a high customer churn. An analytics study indicated the profile of these customers to be young adults, who earns a monthly income of less than R4000, and who do not own any fixed assets.

Vir elkeen van die volgende analitika gebruiksgevalle, identifiseer of dit 'n voorbeeld is van beskrywende analitika, diagnostiese analitika, voorspellende analitika, of voorskriftelike analitika.

(a) Vanaf 'n data stel wat deur 'n gesondheidsdiens verskaffer beskikbaar gestel is, identifiseer pasiënte wat klinies oorgewig is. [1]

(b) Vir daardie pasiënte wat as oorgewig aangedui is, stel 'n behandelingsplan voor met verywsing na diabetes in LDL cholesterol. [1]

(c) 'n Selfoon kommunikasie verskaffer het baie kliënt kansellasies ondervind. 'n Analitika studie het die profiel van hierdie kliënte aangedui as jong volwassenes met 'n maandelikse inkomste van minder as R4000 en wie geen vaste bates besit nie. [1]

Section B: /*Afdeling B* **[15]**

This section covers the machine learning for data analytics topic of the study material./*Hierdie afdeling dek die masjienleer vir data analitika onderwerp van die studie materiaal.*

Question 1: /*Vraag 1* **[2]**

What is predictive data analytics?

Wat is voorspellende data analitika?

Question 2: /*Vraag 2* **[5]**

This question refers to the credit score data provided in Figure 1 and the two models below, induced from this data set./*Hierdie vraag verwys na die krediet telling data wat in Figuur 1 verskaf word, en die twee modelle hieronder, wat vanaf hierdie data geskep is.*

ID	Occupation	Age	Loan-Salary Ratio	Outcome
1	industrial	39	3.40	default
2	industrial	22	4.02	default
3	professional	30	2.7 0	repay
4	professional	27	3.32	default
5	professional	40	2.04	repay
6	professional	50	6.95	default
7	industrial	27	3.00	repay
8	industrial	33	2.60	repay
9	industrial	30	4.5 0	default
10	professional	45	2.78	repay

Figure 1: Credit Score Data/*Krediet Telling Data*

Model 1

if Loan-Salary Ratio > 3.00 **then**

Outcome = *default*

else

Outcome = *repay*

Model 2

if Age= 50 **then**

Outcome = *default*

else if Age= 39 **then**

Outcome = *default*

else if Age= 30 **and** Occupation = *industrial* **then**

Outcome = *default*

else if Age= 27 **and** Occupation = *professional* **then**

Outcome = *default*

else

Outcome = *repay*

(a) Is *Model 1* consistent with the data set? Motivate your answer.

(b) Which of these two models do you think will generalise better to instances not contained in the dataset? Motivate your answer.

(a) Is *Model 1* konsekwent met die data stel? Motiveer jou antwoord. [2]

(b) Watter een van hierdie twee modelle dink jy sal beter veralgemeen na gevalle wat nie in die data stel voorkom nie? Motiveer jou antwoord. [3]

Question 3: /*Vraag 3* **[6]**

With reference to machine learning,

(a) what is meant by the term inductive bias?

(b) what can go wrong when an inapproriate inductive bias is used? Explain your answer. You may use illustrations in your answer.

Met verwysing na masjienleer,

(a) wat word met die term induktiewe vooroordeel bedoel? [2]

(b) wat kan verkeerd gaan wanneer 'n onvanpaste induktiewe vooroordeel gebruik word? Verduidelik jou antwoord. Jy kan illustrasies in jou antwoord gebruik. [4]

Question 4: /*Vraag 4* **[2]**

What is the main difference between supervised and unsupervised learning?

Wat is die hoof verskil tussen toesighoudende en nie-toesighoudende leer?

Section C: /*Afdeling C* **[10]**

This section covers the data-to-insights-to-decisions topic of the study material./*Hierdie afdeling dek die data-na-insigte-na-besluite onderwerp van die studie materiaal.*

Question 1: /*Vraag 1* **[6]**

This question refers to the insurance policy data provided in Figure 2./*Hierdie vraag verwys na die versekeringspolis data wat in Figuur 2 verskaf word.*

ID	OCCUPATION	GENDER	AGE	MOTOR VALUE	POLICY TYPE	PREF CHANNEL
1	lab tech	female	43	42,632	planC	sms
2	farmhand	female	57	22,096	planA	phone
3	biophysicist	male	21	27,221	planA	phone
4	sheriff	female	47	21,460	planB	phone
5	painter	male	55	13,976	planC	phone
6	manager	male	19	4,866	planA	email
7	geologist	male	51	12,759	planC	phone
8	messenger	male	49	15,672	planB	phone
9	nurse	female	18	16,399	planC	sms
10	fire inspector	male	47	14,767	planC	email

Figure 2: Insurance Data/*Versekeringsdata*

(a) Identify which of the features are ordinal, nominal, numeric, and binary.

(b) Which one of the features will you definitely remove prior to building a predictive model? Motivate your answer.

(a) Identifiseer watter van hierdie eienskappe is ordinaal, nominaal, numeries, en binêr. [4]

(b) Watter een van die eienskappe sal jy beslis verwyder alvorens 'n voorspellende model gebou word? Motiveer jou antwoord. [2]

Question 2: /*Vraag 2* [4]

Discuss why feature selection is important with reference to data-driven predictive models.

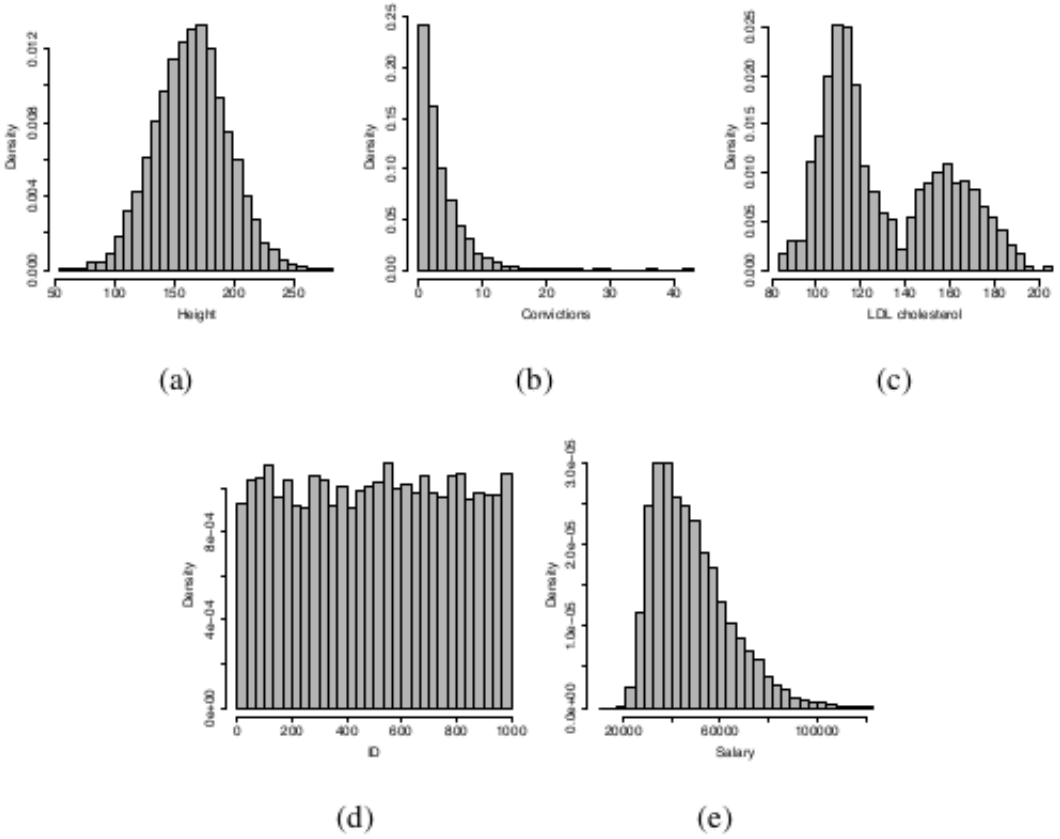
Bespreek hoekom eienskap seleksie so belangrik is in die konteks van data-gedrewe voorspellende modelle.

Section D: /*Afdeling D* [42]

This section covers the data exploration topic of the study material./*Hierdie afdeling dek die data verkenning onderwerp van die studie materiaal.*

Question 1: /*Vraag 1* [4]

Consider the following distributions for some arbitrary features, and answer the questions that follow./*Beskou die volgende verdelings vir 'n paar arbitrêre eienskappe, en beantwoord die vrae wat volg.*



(a) (b) (c)

(d) (e)

(a) Which of these distributions may indicate a feature with outliers? Motivate your answer.

(b) Which one of these distributions indicate a feature that should be considered for removal? Motivate your answer.

(a) Watter een van hierdie verdelings dui 'n eienskap aan met moontlike uitskieters? Motiveer jou antwoord. [2]

(b) Watter een van hierdie verdelings dui aan dat die eienskap eerder verwyder moet word? Motiveer jou antwoord. [2]

Question 2: /*Vraag 2* **[7]**

(a) When binning is used to convert a numeric feature into a categorical feature, what are the consequences of too few bins versus too many bins?

(b) When converting an ordinal feature into a numeric feature, which of one-hot encoding or dummying encoding should be used? Motivate your answer.

(a) Wanneer saamverpakkingsafdeling gebruik word om 'n numeriese eienskap na 'n kategoriese eienskap om te skakel, wat is die gevolge van te min dromme teenoor te veel dromme? [5]

(b) Wanneer 'n ordinale eienskap na 'n numeriese eienskap omgeskakel moet word, watter van een-warm enkodering of dummy enkodering moet gebruik word? Motiveer jou antwoord. [2]

Question 3: /*Vraag 3* **[21]**

This question refers to the medical condition described below. Provided are a description of the problem and the features, part of the ABT, the data quality report and some visualizations. Answer each of the questions that follow./*Hierdie vraag verwys na die mediese kondisie wat hieronder beskryf word. Gegee is 'n beskrywing van die probleem en die eienskappe, 'n deel van die ABT, die data kwaliteit verslag, en 'n paar visualiserings. Beantwoord elk van die vrae wat hierop volg.*

Problem Description/*Probleem Beskrywing* Tachycardia is a condition that causes the heart to beat faster than normal at rest. The occurrence of tachycardia can have serious implications including increased risk of stroke or sudden cardiac arrest. An analytics consultant has been hired by a major hospital to build a predictive model that predicts the likelihood that a patient at a heart disease clinic will suffer from tachycardia in the month following a visit to the clinic. The hospital will use this model to make predictions for each patient when they visit the clinic and offer increased monitoring for those deemed to be at risk./*Tachycardia is 'n kondisie wat veroorsaak dat die hart vinniger as normaal klop in rustende toestand. Die voorkoms van tachycardia kan ernstige gevolge inhou, insluitent 'n risiko vir beroerte of 'n skielike hartaanval. 'n Analitika konsultant is deur 'n groot hospitaal gehuur om 'n voorspellende model te bou wat gebruik sal word om die waarskynlikheid dat pasiënte van 'n hart kliniek aan trachycardia sal ly in die maand wat volg na 'n besoek aan die kliniek, te bepaal. Die hospitaal sal hierdie model gebruik om vir elke pasiënt wat die hospitaal besoek, voorspellings te maak en om addisionele monitering vir risiko pasiënte an te bied.*

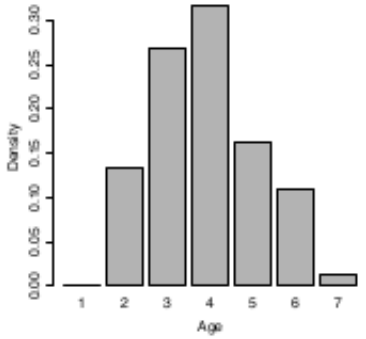
Feature Description/*Eienskap Beskrywing* The features are described below:/*Die eienskappe wor hieronder beskryf:*

- AGE: The patient's age
- GENDER: The patient's gender (*male* or *female*)
- WEIGHT: The patient's weight
- HEIGHT: The patient's height
- BMI: The patient's body mass index (BMI) which is calculated as $\frac{weight}{height^2}$ where weight is measured in kilograms and height in meters.
- SYS. B.P.: The patient's systolic blood pressure
- DIA. B.P.: The patient's diastolic blood pressure
- HEART RATE: The patient's heart rate
- H.R. DIFF.: The difference between the patient's heart rate at this visit and at their last visit to the clinic
- PREV. TACHY.: Has the patient suffered from tachycardia before?
- TACHYCARDIA: Is the patient at high risk of suffering from tachycardia in the next month?

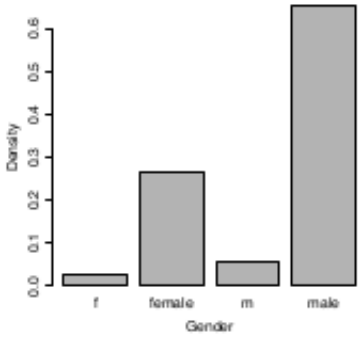
Analytics Base Table/*Analitika Basis Tabel* The following table contains an extract from the ABT; the full ABT contains 2440 instances./*Die volgende tabel bevat 'n uittreksel vanuit die ABT; die volle ABT bevat 2440 gevalle.*

ID	AGE	GENDER	WEIGHT	HEIGHT	BMI	SYS. B.P.	DIA. B.P.	HEART RATE	H.R. DIFF.	PREV. TACHY.	TACHYCARDIA
1	6	male	78	165	28.65	161	97	143			true
2	5	m	117	171	40.01	216	143	162	17	true	true
		⋮			⋮			⋮			
143	5	male	108	1.88	305,568.13	139	99	84	21	false	true
144	4	male	107	183	31.95	1,144	90	94	-8	false	true
		⋮			⋮			⋮			
1,158	6	female	92	1.71	314,626.72	111	75	75	-5		false
1,159	3	female	151	1.59	596,495.39	124	91	115	23	true	true
		⋮			⋮			⋮			
1,702	3	male	86	193	23.09	138	81	83		false	false
1,703	6	f	73	166	26.49	134	86	84	-4		false
		⋮			⋮			⋮			

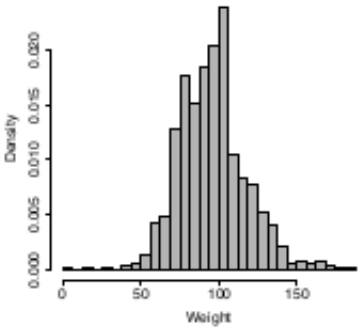
Visualizations/*Visualiserings* The following figures visualize some of the features./*Die volgende figure visualiseer van die eienskappe.*



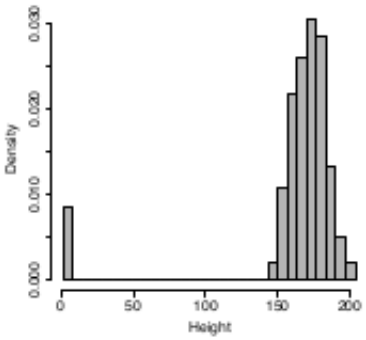
AGE



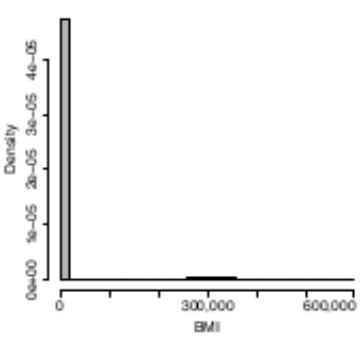
GENDER



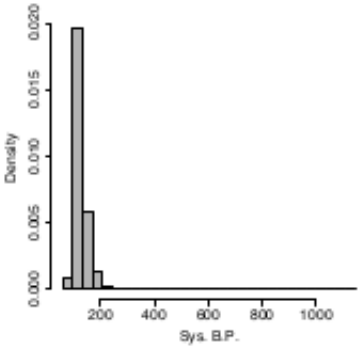
WEIGHT



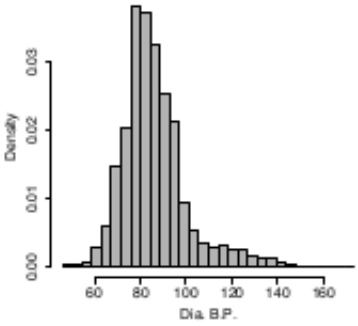
HEIGHT



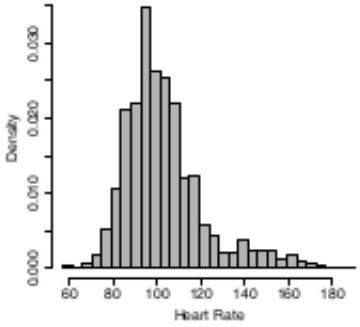
BMI



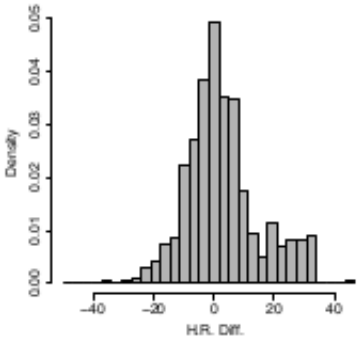
SYS. B.P.



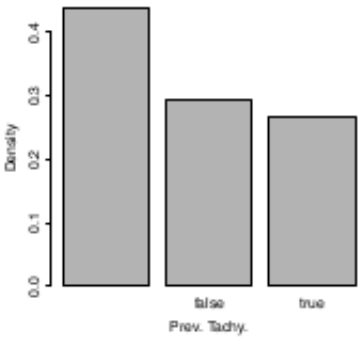
DIA. B.P.



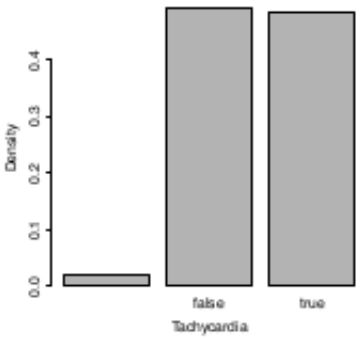
HEART RATE



H.R DIFF.



PREV. TACHY.



TACHYCARDIA

Data Quality Report/*Data Kwaliteit Verslag* The consultant generated the following data quality report from the ABT./*Die konsultant het die volgende data kwaliteit verslag vanaf die ABT geskep.*

Feature	Count	% Miss.	Card.	Mode	Mode Freq.	Mode %	2nd Mode	2nd Mode Freq.	2nd Mode %
GENDER	2,440	0.00	4	male	1,591.00	65.20	female	647.00	26.52
PREV. TACHY.	2,440	44.02	3	false	714.00	52.27	true	652.00	47.73
TACHYCARDIA	2,440	2.01	3	false	1,205.00	50.40	true	1,186.00	49.60

Feature	Count	% Miss.	Card.	Min.	1st Qrt.	Mean	Median	3rd Qrt.	Max.	Std. Dev.
AGE	2,440	0.00	7	1.00	3.00	3.88	4.00	5.00	7.00	1.22
WEIGHT	2,440	0.00	174	0.00	81.00	95.70	95.00	107.00	187.20	20.89
HEIGHT	2,440	0.00	109	1.47	162.00	162.21	171.50	179.00	204.00	41.06
BMI	2,440	0.00	1,385	0.00	27.64	18,523.40	32.02	38.57	596,495.39	77,068.75
SYS .B.P.	2,440	0.00	149	62.00	115.00	127.84	124.00	135.00	1,144.00	29.11
DIA. B.P.	2,440	0.00	109	46.00	77.00	86.34	84.00	92.00	173.60	14.25
HEART RATE	2,440	0.00	119	57.00	91.75	103.28	100.00	110.00	190.40	18.21
H.R. DIFF.	2,440	13.03	78	-50.00	-4.00	3.00	1.00	8.00	47.00	12.38

Discuss the data quality report in terms of the aspects listed below. For each data quality issue, identify all of the features which exhibit that quality issue. Provide a motivation for each of the identified quality issues. Propose a plan for each of the data quality issues, and provide a motivation for each of your plans.

(a) Missing values

(b) Irregular cardinality

(c) Outliers

Answer the following question with reference to the target feature:

(d) Would you suggest that either under-sampling or over-samling be applied to the target feature? Motivate your answer.

Bespreek die data kwaliteit verslag met verwysing na die aspekte wat hieronder gelys word. Vir elkeen van die kwalitietsprobleme, identifiseer all die eienskappe wat daardie probleem toon. Gee 'n motivering vir elk van die geïdentifiseerde kwaliteitsprobleme. Stel 'n plan voor vir elk van die die data kwaliteitsprobleme, en gee 'n motivering vir elk van jou planne.

(a) Verlore waardes [6]

(b) Onreëlmatige kardinaliteit [6]

(c) Uitskieters [6]

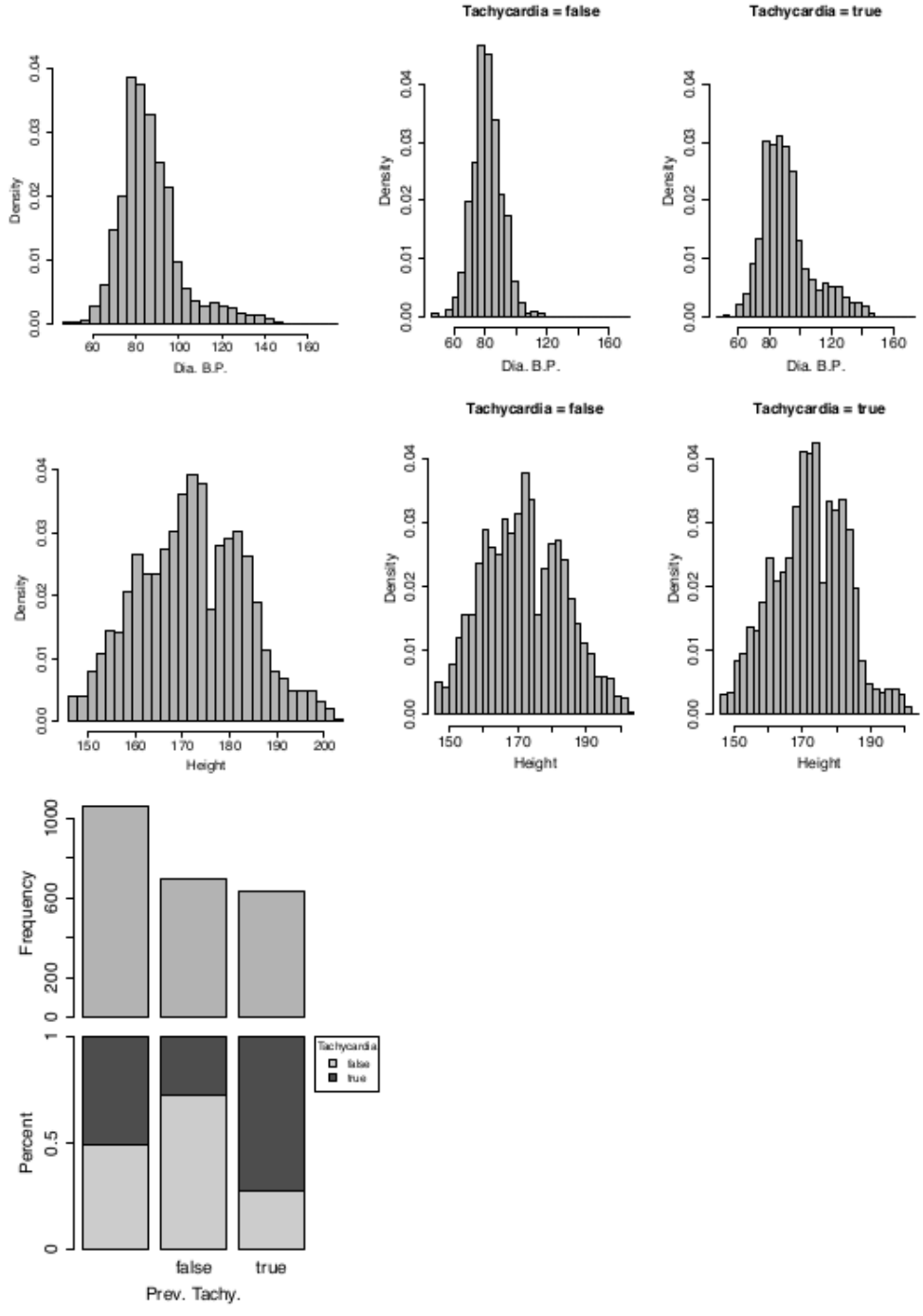
Beantwoord die volgende vraag met verwysing na die teiken eienskap:

(d) Sou jy voorstel dat enige onder-steekproefneming of oor-steekproefneming op die teiken eienskap toegepas moet word? [3]

Question 4: /*Vraag 4* [10]

Assume now that all of the identified data quality issues with respect to the target feature have been corrected. The images below visualize the relationships between some descriptive features and the target feature. For each visualization, three plots are given: a plot of the distribution of the descriptive feature values in the full dataset, and plots showing the distribution of the descriptive feature values for each level of the target. Answer the questions that follow./*Aanvaar nou dat al die data kwaliteitsprobleme ten opsigte van die teiken eienskap gekorrigeer is. Die figure hieronder visualiseer die verwantskappe tussen sommige beskrywende eienskappe en die teiken eienskap. Vir elke visualisering word drie figure gegee: een van die verdeling van die waardes van die beskrywende eienskap oor*

die hele data stel, en dan een figuur vir die verdeling van die beskrywende eienskap se waardes vir elke vlak van die teiken figuur. Beantwoord die vrae wat volg.



(a) Is there a relationship between feature DIA. B.P. and the target feature TACHYCARDIA? Motivate your answer.

(b) From this relationship, which of the following patient groups are more likely to suffer from tachycardia: patiets with higher diastolic blood pressure or patients with lower blood pressure values? Motivate your answer.

(c) True or false: The Height feature should be removed. Motivate your answer.

(d) With reference to the relationship between PREV. TACHY and TACHYCARDIA, do you think that the large number of missing values for PREV. TACHY presents a problem, and that this descriptive feature should be removed? Motivate your answer.

(e) What information can you derive about the relationship between PREV. TACHY and TACHYCARDIA

(a) Is daar 'n verwantskap tussen eienskap DIA. B.P. en die teiken eienskap TACHYCARDIA? Motiveer jou antwoord. [2]

(b) Vanuit hierdie verwantskap, watter een van die volgende pasiënt groepe is meer waarskynlik om te ly aan tachycardia: pasiënte met hoër waardes vir diastoliese bloeddruk, of pasiënte met laer bloeddruk waardes? Motiveer jou antwoord. [2]

(c) Waar of vals: Die Height eienskap moet verwyder word. Motiveer jou antwoord. [2]

(d) Met verwysing na die verwantskap tussen PREV. TACHY en TACHYCARDIA, dink jy dat die groot aantal verlore waardes vir PREV. TACHY problematies is, en dat hierdie beskrywende eienskap verwyder moet word? Motiveer jou antwoord. [2]

(e) Watter inligting kan jy aflei van die verwantskap tussen PREV. TACHY en TACHYCARDIA. [2]